Juan Antonio Montalbán Vidal

MATHEMATICS | SCIENTIFIC MACHINE LEARNING

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MSc candidate in Mathematics working on physics-informed neural networks (PINNs). Training in statistics, optimisation and mathematical epidemiology, with experience in formalisation, scientific computing and applied machine learning.

EDUCATION

MSc in Mathematics | *University of Seville, Spain* Oct 2025 - Nov 2026

- Specialisation in Analysis, Statistics and Operations Research. Current average: **9.1**/**10**; thesis ongoing.
- **Honours (Matrícula de Honor) in Optimisation**.
- Prior graduate study at the **University of Bonn**, Germany (Sep 2024 - Mar 2025), before transfer to Seville.

Master's in Advanced Data Analytics, Big Data and AI | *Instituto de Estudios Cajasol, Spain* Oct 2025 - Jun 2026

- Grade: **9**/**10**. Fully funded by the *Beca de Excelencia*, awarded by Fundación Cajasol on academic merit.
- Thesis: *Data Analysis and Predictions in Sports - the case of tennis*; see Selected Applied Project.

***Erasmus+* Exchange in Mathematics** | *University of Warwick, UK* Sep 2023 - Jun 2024

- Exchange-year grade: **First Class Honours**. Modules included Mathematics of Machine Learning, Linear Statistical Modelling with R and Commutative Algebra.
- **Epidemiology by Example:** SIR/SEIR models, dynamical systems and stability analysis, deterministic and stochastic modelling, frequentist and Bayesian inference (MLE/MCMC), and numerical simulation in R.

BSc in Mathematics | *University of Seville, Spain* Sep 2020 - Jul 2024

- Grade: **8.59**/**10**; **top 5%** of the cohort. Honours in Algebraic Structures, Probability Theory and Complex Analysis.

RESEARCH EXPERIENCE & MATHEMATICAL PROJECTS

MSc Thesis: Physics-Informed Neural Networks | *University of Seville* Ongoing

- *Computer-Assisted Analysis: the case of PINNs and applications to the De Gregorio model*. Established a new result on the linear instability of the De Gregorio model. Used PINNs to heuristically identify a candidate eigenvalue and guide the search, followed by analytical validation of its existence.
- Literature review of PINN approaches to fluid dynamics; comparison of the strengths and limitations of PINNs and spectral methods.
- Supervisor: **Francisco Javier Torres de Lizaur**. Manuscript in preparation; defence scheduled for November 2026.

BSc Thesis: Mathematical Physics | *University of Seville* 2024

- *Mathematical Foundations of the Quantum Harmonic Oscillator.* Mark: **10**/**10**.
  Supervisor: Prof. Dr. Renato Álvarez Nodarse.

Formalisation in Lean 4 | *Collaborative mathematical projects*

- **Bonn:** *Construction of the Winding Number of Complex Curves*.
- **Warwick:** *Proof of Cauchy's Theorem on a triangle using Goursat's method*.

Student Research Intern | *Mathematical Analysis Department, University of Seville* Sep 2022 - Jun 2023

- Conducted an undergraduate research project on elliptic functions and their applications to modern mathematics.

TECHNICAL WRITING & OPEN-SOURCE CONTRIBUTIONS

[1] Elías Zamora Sillero, Joe Olson, Juan A. Cabeza Souza, Agata Ferretti, Milan Jobard-Hoffmann, **Juan A. Montalbán Vidal** and Andrea Greggo. *Building a governed conversational BI agent: Lessons from a retail PoC*. Technical article, The AI Alliance, June 2026.

[2] **Contributor**, The AI Alliance: *conversational_bi* (2026). Open-source reference implementation for governed natural-language data querying, with ontology mapping, validation and confidence-aware responses.

PROFESSIONAL EXPERIENCE

Data & AI Engineering Intern | *Technology Department, Sevilla FC, Seville, Spain* Apr 2026 - Jun 2026

- **Football analytics and conversational scouting**
  - Built a football scouting assistant for natural-language exploration of performance and private market data, using the conversational BI framework developed by an international team within The AI Alliance.
  - Implemented player search, similarity analysis and rankings; connected requests to Azure SQL through entity disambiguation, reusable queries and explicit handling of missing data and unsupported filters.
- **Governed Conversational BI and The AI Alliance**
  - Co-developed a conversational BI proof of concept with researchers from **IBM Research** within The AI Alliance, combining domain validation, ontology mapping and explicit confidence levels for natural-language SQL queries.
  - Owned the ontology layer: designed its hierarchy and search index and implemented LLM-assisted mapping from user terminology to structured data concepts.
  - Contributed to the open-source demonstrator, coordinated GitHub contributions across institutions and co-authored the project's technical article.

Data & AI Engineer | *Keller Williams GSI, Seville, Spain* Jul 2025 - Nov 2025

- Developed Python agent workflows with LangChain, LangGraph and FastMCP to identify real-estate opportunities and match clients with suitable properties.

SELECTED APPLIED PROJECT

ATP Tennis Analytics & AI | *Master's project, Instituto de Estudios Cajasol* 2026

- Engineered **86 features from 28,000+ ATP matches** for XGBoost prediction using prior-match information, including surface-specific Elo and recent performance statistics; built Python ETL and relational data models.
- Compared tournament predictions with **Elo baselines**, analysing probability calibration, performance by round and limitations for players with sparse match histories.
- Explored player profiles through **PCA and k-means clustering**, created a six-page Power BI dashboard and developed an AI assistant with 13 MCP tools for data exploration and access to model predictions.

TECHNICAL SKILLS & LANGUAGES

- **Scientific computing:** Python, R, MATLAB, JAX, PyTorch; AMPL for optimisation modelling.
- **Machine learning & analysis:** pandas, scikit-learn, XGBoost, Power BI.
- **Data & software:** SQL, Azure SQL, Oracle Cloud, DuckDB, Parquet, Git/GitHub, Haskell.
- **AI engineering:** LLM integration, FastMCP, LangChain, LangGraph.
- **Formalisation & documentation:** Lean 4, LaTeX.
- **Languages:** Spanish (native), **English (C1 - Cambridge Advanced)**, French (lower intermediate), German (A1).
- **Certifications:** AZ-900: Microsoft Azure Fundamentals; Oracle Cloud Infrastructure Certified Foundations Associate.

HONOURS & ACCOLADES

- Erasmus+ Scholarship for the 2023-2024 academic year at Warwick Mathematics Institute.
- Banco Santander Erasmus+ Scholarship, awarded on academic merit.

VOLUNTEERING

ECM Volunteer *9th European Congress of Mathematics, Seville, Spain* Jul 2024

- Helped organise the congress, including welcoming participants and preparing conference sessions.